

Numerical solver 3-link IK

$$J = \begin{bmatrix} \frac{\partial x}{\partial \theta_1} & \frac{\partial x}{\partial \theta_2} & \frac{\partial x}{\partial \theta_3} \\ \frac{\partial y}{\partial \theta_1} & \frac{\partial y}{\partial \theta_2} & \frac{\partial y}{\partial \theta_3} \end{bmatrix} \quad f(\theta_1, \theta_2, \theta_3) = \begin{pmatrix} f_1(\theta_1, \theta_2, \theta_3) \\ f_2(\theta_1, \theta_2, \theta_3) \end{pmatrix}$$

$$x = L_1 \cos(\theta_1) + L_2 \cos(\theta_1 + \theta_2) + L_3 \cos(\theta_1 + \theta_2 + \theta_3)$$

$$y = L_1 \sin(\theta_1) + L_2 \sin(\theta_1 + \theta_2) + L_3 \sin(\theta_1 + \theta_2 + \theta_3)$$

$$\frac{\partial x}{\partial \theta_1} = -L_1 \sin(\theta_1) - L_2 \sin(\theta_1 + \theta_2) - L_3 \sin(\theta_1 + \theta_2 + \theta_3)$$

$$\frac{\partial x}{\partial \theta_2} = -L_2 \sin(\theta_1 + \theta_2) - L_3 \sin(\theta_1 + \theta_2 + \theta_3)$$

$$\frac{\partial x}{\partial \theta_3} = -L_3 \sin(\theta_1 + \theta_2 + \theta_3)$$

$$\frac{\partial y}{\partial \theta_1} = L_1 \cos(\theta_1) + L_2 \cos(\theta_1 + \theta_2) + L_3 \cos(\theta_1 + \theta_2 + \theta_3)$$

$$\frac{\partial y}{\partial \theta_2} = L_2 \cos(\theta_1 + \theta_2) + L_3 \cos(\theta_1 + \theta_2 + \theta_3)$$

$$\frac{\partial y}{\partial \theta_3} = L_3 \cos(\theta_1 + \theta_2 + \theta_3)$$

Same idea as numerical solver for 2-link.

$$\Delta p = \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix}, \quad J = \begin{bmatrix} \frac{\partial x}{\partial \theta_1} & \frac{\partial x}{\partial \theta_2} & \frac{\partial x}{\partial \theta_3} \\ \frac{\partial y}{\partial \theta_1} & \frac{\partial y}{\partial \theta_2} & \frac{\partial y}{\partial \theta_3} \end{bmatrix} \quad \theta = \begin{bmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \end{bmatrix}$$

$$\Delta p = J \Delta \theta \Rightarrow \Delta \theta = J^+ \Delta p \quad \text{error} \approx \Delta p \Rightarrow \Delta \theta \approx J^+ \text{error}$$

The pseudoinverse Jacobian method computes joint updates using

$$\Delta\theta \approx J^+ \text{ error}$$

where error is the vector from the current end-effector position to the target.

For the 3-link arm, the Jacobian is 2×3 because the task is 2D position control but there are 3 joint angles. This means the arm is redundant, and there are many possible joint configurations that can reach the same target.

One issue with the pseudoinverse method is that it can become unstable near singular configurations, where the arm loses useful movement directions or the Jacobian becomes poorly conditioned. In those cases, the pseudoinverse can produce very large joint updates.

To improve stability, I also implemented damped least squares. This modifies the update rule by adding a damping term, which reduces extreme updates:

$$\Delta\theta \approx J^T(JJ^T + \lambda^2 I)^{-1} \text{ error}$$

The damping parameter λ controls the tradeoff between accuracy and stability. A small λ behaves similarly to the pseudoinverse, while a larger λ makes the solver more stable but slower or less precise.

Also known as Levenberg - Marquardt method.